

Employment

Graduate Research Assistant NC State University April 2025 – Present

- Deployed containerized genomic platform on GCP Cloud Run (Docker, Cloud Build CI) with JBrowse 2 genome browser; enabled 100+ researchers to visualize 500GB+ datasets by eliminating on-premise infrastructure.
- Improved ML model accuracy by 12% via grid-search hyperparameter sweeps and sliding-window feature engineering on multivariate biological sensor time-series.

Product Engineer Intern Isomer AI May 2025 – May 2026

- Shipped 10+ production AI features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST APIs; 923 commits via GitHub Actions CI/CD across 7 deployments.
- Owned PostgreSQL, Redis, Auth0 RBAC, Terraform, Docker, and Sentry stack; triaged and resolved production incidents by tracing failures across async microservices.

Software Developer Intern Mphasis (HP Company) Jun 2024 – Aug 2024

- Improved document processing accuracy by 25% by building OCR + NLP classification pipelines that parsed unstructured health claim documents into structured fields for downstream automation.
- Automated AWS and GCP releases via GitHub Actions CI/CD; configured Grafana dashboards for observability and led red-teaming to surface model reliability gaps pre-production.

AI/ML Intern Capgemini Jun 2023 – Aug 2023

- Reduced ticket resolution time by 15% by training an NLP multi-class classifier that auto-routed support tickets to the correct team based on predicted category and urgency.
- Cut ticket-processing latency by 20% by replacing synchronous DB polling with Kafka async ingestion feeding Spark micro-batch jobs across partitioned Hadoop HDFS.

Education

Raleigh, NC North Carolina State University Aug 2024 – May 2026

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.

Mumbai, India Mumbai University Aug 2020 – May 2024

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.

Projects

PulseRank Real-Time Leaderboard & Activity Feed 2026

- Built event-driven leaderboard in Java/Spring Boot and TypeScript/NestJS: MySQL transactional writes with idempotency-key deduplication and outbox pattern, MongoDB TTL projections, Redis sorted sets for sub-millisecond reads.
- Validated multi-store consistency with Testcontainers integration tests against live DB containers; benchmarked hot-path latency with k6 and enforced test gates in GitHub Actions CI.

VectorLift Semantic Search & Ranking Engine 2026

- Built Kafka-driven corpus ingestion pipeline into Elasticsearch and Qdrant; served semantic search via FastAPI with per-stage latency tracking across 6 configurable pipeline configurations.
- Improved retrieval quality by fine-tuning bi-encoder (MS MARCO) and cross-encoder re-ranker (MiniLM); evaluated with NDCG@10, MRR@10, and paired bootstrap significance testing.

Languages and Technologies

- **Languages:** Java, Python, TypeScript, JavaScript, SQL, Bash, C
- **Backend & APIs:** Spring Boot, NestJS, FastAPI, REST APIs, event-driven architecture, outbox pattern, Kafka
- **Data & Storage:** PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Hadoop, Spark, OpenTelemetry
- **Infrastructure:** Kubernetes, Docker, Helm, ArgoCD, AWS, GCP, Terraform, GitHub Actions, Prometheus, Grafana

Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog